

bpf man

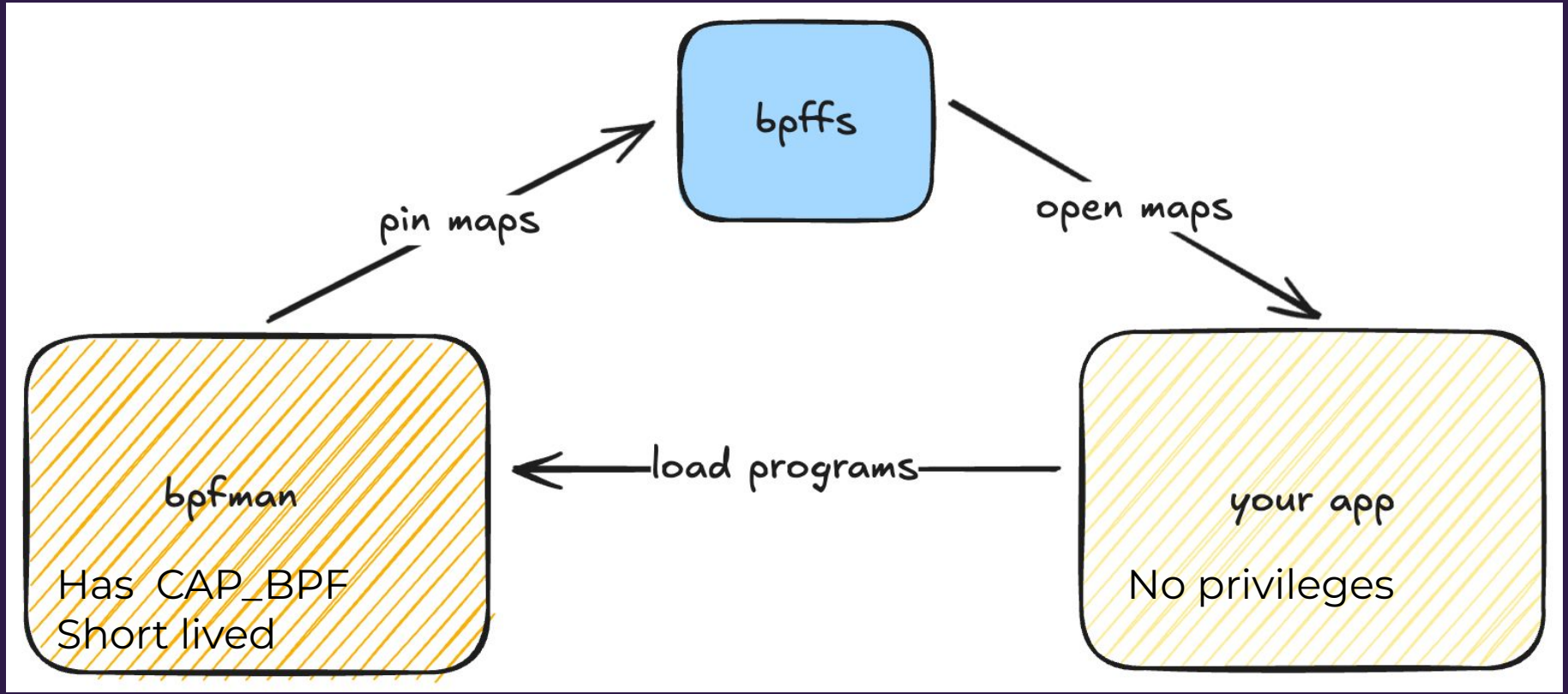
What is bpfman?

- A suite of tools for deploying, managing and observing eBPF
- Works on Linux
- Works on Kubernetes

What problems does bpfman solve?

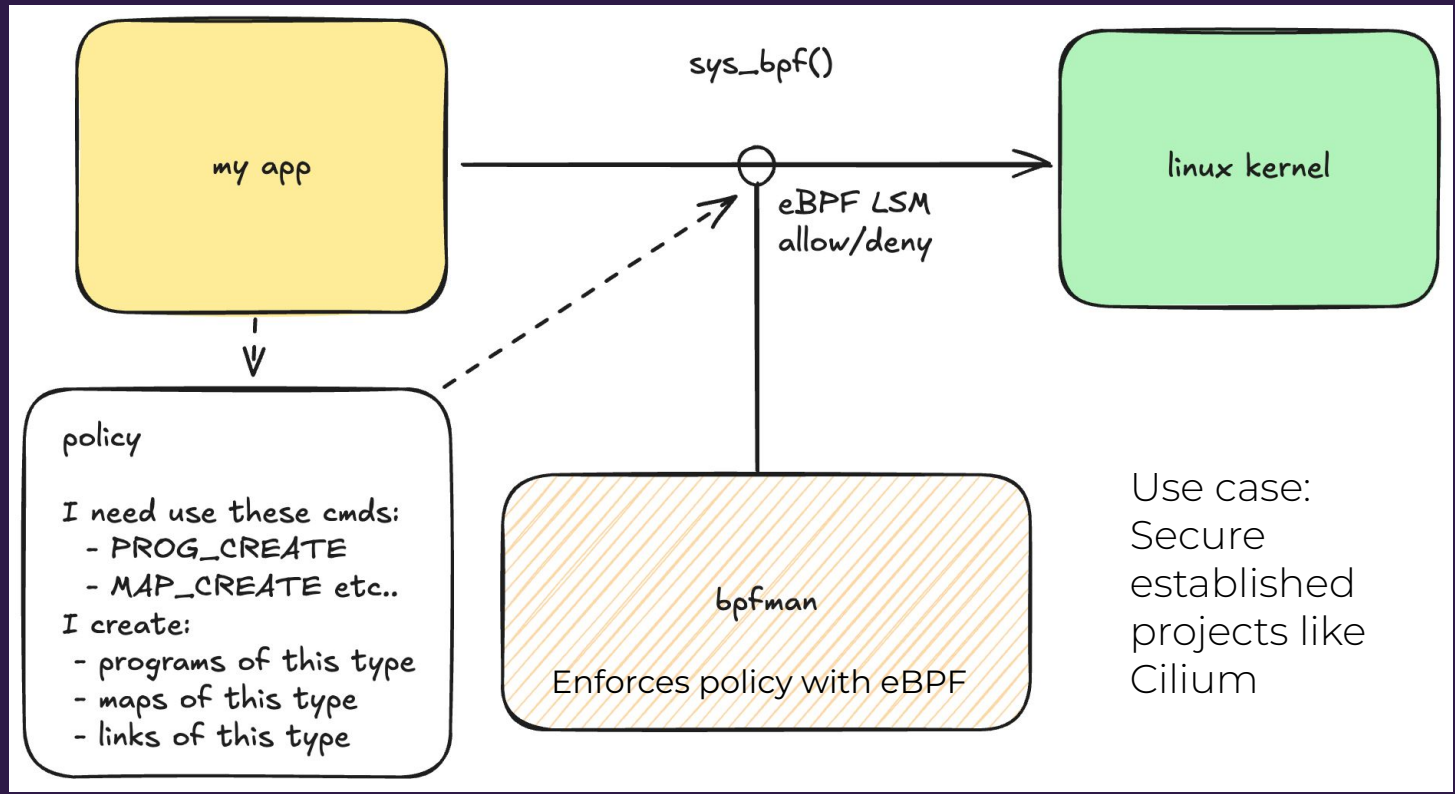
- No privileges required for userspace apps/pods (no CAP_BPF)
  - Policing of CAP_BPF granted to apps/pods COMING SOON!
- Daemon-less design - like podman, but for eBPF
- Easy packaging and signing of eBPF code using OCI images
- Declarative configuration on Kubernetes (Custom Resources)
- Observability of eBPF usage

No CAP_BPF?


bpffs
pin maps
open maps
bpfman
Has CAP_BPF
Short lived
load programs
your app
No privileges

CAP_BPF Policing!

COMING SOON!



Use case: Secure established projects like Cilium

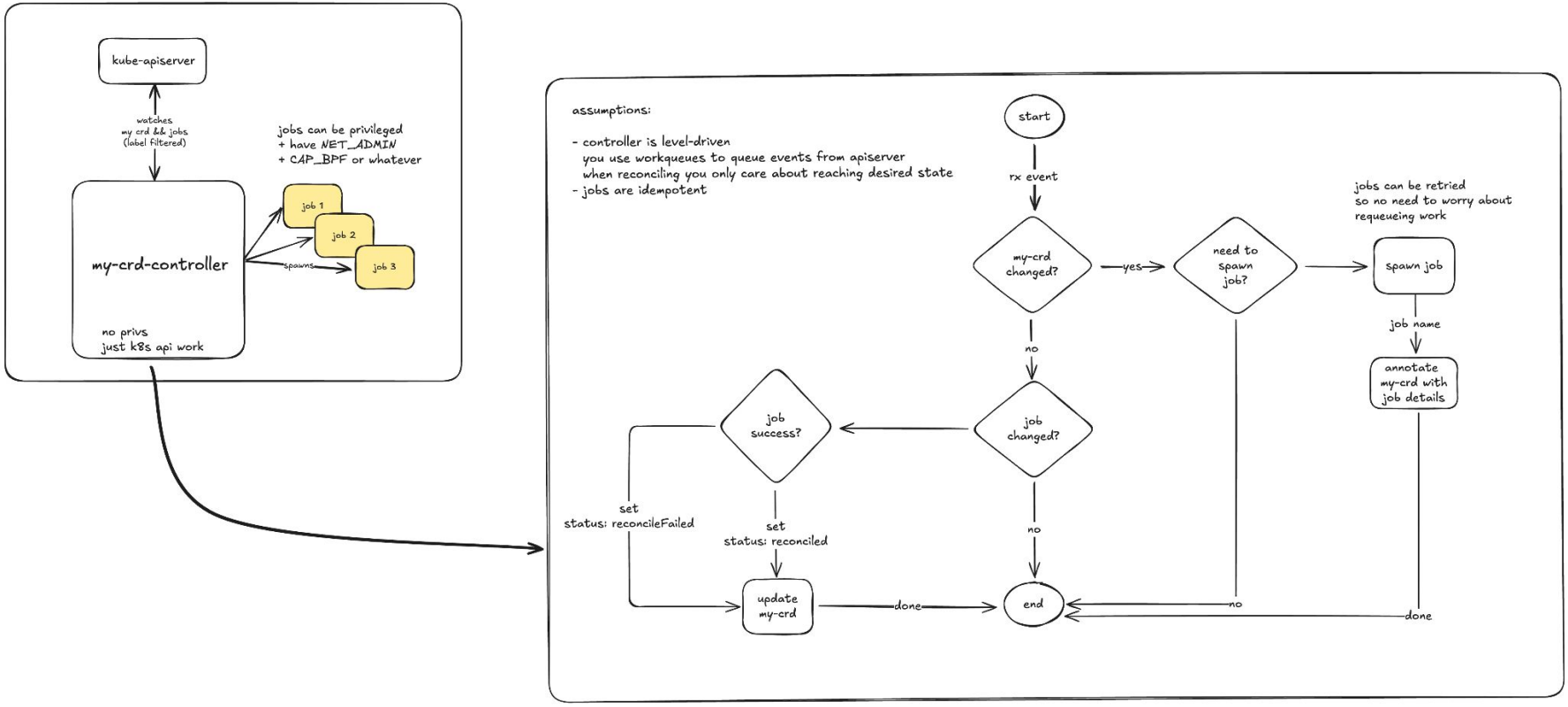
Daemon-less (Linux Only)

- Like podman!

- Loads your eBPF code and exits avoiding keeping elevated privileges for a long time

Daemon(set)-less (K8s only)

COMING SOON!



OCI images

- bpfman build - creates multi-arch manifests with containing the right eBPF bytecode for the target architecture
- Push images to any OCI-compatible registry
- Use cosign to sign your container images



- bpfman can verify signatures!
  - Finer-grained attestations

COMING SOON!

Declarative Configuration (K8s) - Kernel space

```yaml
apiVersion: bpfman.io/v1alpha1
kind: ClusterBpfApplication
metadata:
  labels:
    app.kubernetes.io/name: my-app
  name: my-app
spec:
  nodeSelector: {}
  byteCode:
    image:
      url: quay.io/example/example-ks:latest
      imagePullPolicy: Always
```

```yaml
programs:
    - name: kprobe_counter
      type: KProbe
      kprobe:
        links:
          - function: try_to_wake_up
            offset: 0
    - name: tracepoint_kill_recorder
      type: TracePoint
      tracepoint:
        links:
          - name: syscalls/sys_enter_kill
```

Declarative Configuration (K8s) - User space

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
  name: my-app-ds
spec:
  template:
    spec:
      containers:
        - name: my-app
          image: quay.io/example/example-us:latest
          imagePullPolicy: Always
          capabilities:
            drop: [ALL]
          volumeMounts:
            - name: my-app-maps
              mountPath: /run/app/maps
              readOnly: true
      volumes:
        - name: my-app-mapps
          csi:
            driver: csi.bpfman.io
            volumeAttributes:
              csi.bpfman.io/program: my_app
              csi.bpfman.io/maps:
                  kprobe_stats_map,tracepoint_stats_map
```

Observability!

Deployment via our operator COMING SOON!



Want to learn more?



Read the docs!

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Earn a sticker!

Proudly written in Rust



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